Class 6 Science – Chapter 8: A Journey through the States of Water

इस अध्याय के PDF के आधार पर नीचे सभी दिखाई देने वाले प्रश्नों के उत्तर Simple English में दिए गए हैं। इसके साथ प्रत्येक मुख्य टॉपिक से महत्वपूर्ण और संभावित परीक्षा-प्रश्न, उनके उत्तर, MCQs, Fill in the Blanks और True/False भी शामिल हैं।

08-a-journey-through-the-state-of-water.pdf

1. Observe and Answer (Page 104)

Q1. What does the given image represent?

Answer: The image represents a polluted water body near an industrial area.

Q2. From where are the wastes added to this water body?

Answer: The wastes may come from factories, industries, and nearby human activities.

Q3. Is the water in this water body potable? Give a reason for your answer.

Answer: No, the water is not potable because it appears polluted. It may contain harmful substances and germs.

Q4. Can this water be used for cleaning clothes and washing utensils?

Answer: No, polluted water should not be used for washing clothes and utensils because it may contain harmful chemicals and germs.

2. Topic: Water

Q1. What is water?

Answer: Water is an important natural resource. All living organisms need water to survive.

Q2. Why is water essential for life?

Answer: Water is essential because humans, animals, and plants need it to live and grow.

Q3. What are the common uses of water?

Answer: We use water for drinking, cooking, bathing, washing clothes, cleaning utensils, farming, and many other activities.

Q4. Why is water called an important natural resource?

Answer: Water is called an important natural resource because it supports life and is needed for many daily activities.

Q5. Why is water necessary for plants?

Answer: Plants need water to grow, make food, and remain healthy.

Q6. Why should we not waste water?

Answer: We should not waste water because only a small amount of the total water on Earth is available for our use.

3. Distribution of Water on Earth

Q1. What percentage of the Earth's surface is covered by water?

Answer: About 71% of the Earth's surface is covered by water.

Q2. What percentage of the Earth's total water is freshwater?

Answer: About 2.5% of the Earth's total water is freshwater.

Q3. Where is most of the water on Earth found?

Answer: Most of the water on Earth is found in seas and oceans as salty water.

Q4. Where is freshwater found?

Answer: Freshwater is found in glaciers, ice caps, groundwater, rivers, lakes, and the air.

Q5. Is all freshwater available for our use?

Answer: No. Most freshwater is frozen in glaciers and ice caps. Only a very small amount is easily available for our use.

Q6. Why is the Earth called the Blue Planet?

Answer: The Earth is called the Blue Planet because about 71% of its surface is covered with water, making it appear blue from space.

Q7. Why should we conserve water?

Answer: We should conserve water because the amount of freshwater available for our use is very small.

4. States of Water

Water exists in three states: solid, liquid, and gas.



Q1. What is the solid state of water?

Answer: In the solid state, water exists as ice, snow, or frost. Ice has a fixed shape and a fixed volume.



Q2. What is the liquid state of water?

Answer: In the liquid state, water flows easily. It has a fixed volume but takes the shape of its container.



Q3. What is the gaseous state of water?

Answer: In the gaseous state, water exists as water vapour or steam. It has no fixed shape or volume and spreads in the available space.

Q4. What determines the state of water?

Answer: The way water molecules are arranged and held together determines its state.

Q5. What is the difference between a solid, a liquid, and a gas?

Property	Solid	Liquid	Gas
Shape	Fixed	Takes the shape of its container	No fixed shape
Volume	Fixed	Fixed	No fixed volume
Movement	Molecules remain closely packed	Molecules can move past each other	Molecules move freely
Example	Ice	Water	Water vapour

5. Change in the State of Water

Water can change from one state to another by adding or removing heat energy.

Q1. What is melting?

Answer: Melting is the process in which a solid changes into a liquid on gaining heat. For example, ice changes into water.

Q2. What is freezing or solidification?

Answer: Freezing is the process in which a liquid changes into a solid on losing heat. For example, water changes into ice in a freezer.

Q3. What is vaporisation?

Answer: Vaporisation is the process in which a liquid changes into a gas or vapour.

Q4. What is condensation?

Answer: Condensation is the process in which a gas changes into a liquid on cooling. For example, water droplets form on the inner surface of a lid placed over boiling water.

Q5. What happens when juice is kept in a freezer for a few hours?

Answer: The juice loses heat and changes from a liquid into a solid. It forms an ice lolly or popsicle.

Q6. How can water change from ice into water and then into water vapour?

Answer: Ice gains heat and melts into water. On further heating, the water changes into water vapour through vaporisation.

Q7. Are changes in the state of water reversible?

Answer: Yes. These changes are generally reversible. Ice melts into water, and water can freeze again into ice. Water vapour can condense into liquid water.

Q8. How does candle wax change from solid to liquid and then back to solid?

Answer: Candle wax melts when heated. When it cools, it becomes solid again.

6. Vaporisation: Evaporation and Boiling

Vaporisation occurs in two ways: evaporation and boiling.

Q1. What is evaporation?

Answer: Evaporation is the process in which a liquid changes into vapour from its surface. It can occur at all temperatures.

Q2. What is boiling?

Answer: Boiling is the process in which a liquid changes into vapour on heating to its boiling point.

Q3. What is the difference between evaporation and boiling?

Evaporation	Boiling
It occurs at the surface of a liquid.	It occurs throughout the liquid.
It can occur at all temperatures.	It occurs at the boiling point.
It usually takes place slowly.	It usually takes place rapidly.
Example: Wet clothes drying.	Example: Water boiling in a pan.

Q4. Why does nail paint remover disappear when left open?

Answer: Nail paint remover contains acetone, which evaporates easily. It changes into vapour and mixes with the air.

Q5. Does evaporation occur only when water is heated?

Answer: No. Evaporation occurs at all temperatures, although heating makes it faster.

Q6. What is transpiration?

Answer: Transpiration is the process by which plants lose water in the form of water vapour through their leaves.

Q7. How is transpiration related to evaporation?

Answer: Both processes release water into the air as water vapour. Evaporation occurs from a liquid surface, while transpiration occurs through plants.

7. Condensation

Q1. What is condensation?

Answer: Condensation is the change of a gas or vapour into a liquid when it cools.

Q2. Why do water droplets form on the inner surface of a lid placed over boiling water?

Answer: Water vapour from the boiling water touches the cooler lid, loses heat, and changes into liquid droplets.

Q3. Why do water droplets appear on the outside of a glass containing a cold drink?

Answer: Water vapour in the air touches the cold surface of the glass and condenses into tiny water droplets.

Q4. What is fog, and how is it formed?

Answer: Fog is made of tiny water droplets suspended near the ground. It forms when water vapour in the air cools and condenses.

Q5. Why do we observe dewdrops on plants in the morning?

Answer: Water vapour in the air cools during the night and condenses on cool plant surfaces, forming dewdrops.

Q6. What is humidity?

Answer: Humidity is the amount of water vapour present in the air.

Q7. Why does water sprinkled on rooftops disappear after some time?

Answer: The water absorbs heat from its surroundings and evaporates into water vapour.

8. Rate of Evaporation

Q1. What is the rate of evaporation?

Answer: The rate of evaporation is the speed at which a liquid changes into vapour.

The chapter explains four important factors that affect the rate of evaporation.

A. Surface area

Q2. How does surface area affect evaporation?

Answer: A larger exposed surface area increases the rate of evaporation.

Q3. Why does water in a plate evaporate faster than the same amount of water in a bowl?

Answer: Water in a plate has a larger surface area exposed to air, so it evaporates faster.

B. Temperature

Q4. How does temperature affect evaporation?

Answer: Higher temperature increases the rate of evaporation because the liquid gains more heat energy.

Q5. Why do clothes dry faster on hot, sunny days?

Answer: The higher temperature increases evaporation, so the water in the clothes evaporates faster.

C. Wind

Q6. How does wind affect evaporation?

Answer: Faster wind increases evaporation by carrying water vapour away from the liquid's surface.

Q7. Why do clothes dry faster on a windy day?

Answer: Wind carries away the water vapour around the clothes, allowing more water to evaporate.

D. Humidity

Q8. How does humidity affect evaporation?

Answer: Higher humidity decreases the rate of evaporation because the air already contains more water vapour.

Q9. Why do clothes dry slowly on rainy days?

Answer: The air is usually more humid on rainy days, so evaporation takes place more slowly.

Q10. Name the four factors that affect the rate of evaporation.

Answer: The four factors are:

1. Surface area
2. Temperature
3. Wind speed
4. Humidity

Quick comparison

Factor increases	Effect on evaporation
Surface area	Increases
Temperature	Increases
Wind speed	Increases
Humidity	Decreases

9. Activity: Evaporation in Sunlight and Shade (Page 110)

Q1. What is the aim of this activity?

Answer: The aim is to observe that evaporation is faster in warm conditions.

Q2. What materials are required?

Answer: Two bowls of the same size and water are required.

Q3. How is the activity performed?

Answer: Pour equal amounts of water into both bowls. Keep one bowl in sunlight and the other in the shade. Observe the water levels at regular intervals.

Q4. What is the observation?

Answer: The water level decreases faster in the bowl kept in sunlight.

Q5. What is the conclusion?

Answer: Evaporation is faster at higher temperatures.

10. Cooling Effect of Evaporation

Q1. What is the cooling effect of evaporation?

Answer: When a liquid evaporates, it takes heat from its surroundings. This makes the surroundings cooler.

Q2. Why is water kept in a clay pot (matka) cooler than water kept in a steel container?

Answer: A clay pot has tiny pores. A small amount of water seeps through these pores and evaporates from the outer surface. During evaporation, it takes heat from the pot and the water inside, making the water cooler.

Q3. Why does alcohol in hand sanitiser make our hands feel cool?

Answer: Alcohol evaporates quickly and takes heat from our skin, making our hands feel cool.

Q4. Should we use a clay pot or a stainless-steel container for storing water if we want the water to stay cool? Why?

Answer: We should use a clay pot because evaporation through its pores helps cool the water.

Q5. Why do we feel cool when sweat dries from our skin?

Answer: Sweat evaporates by taking heat from our skin, which produces a cooling effect.

11. How Do Clouds Give Us Rain?

Q1. How does water enter the atmosphere?

Answer: Water enters the atmosphere through evaporation from water bodies and transpiration from plants.

Q2. How are clouds formed?

Answer: Water vapour rises into the atmosphere and cools. It condenses into tiny water droplets, which gather to form clouds.

Q3. What is precipitation?

Answer: Precipitation is the process by which water falls from clouds to the Earth as rain, hail, or snow.

Q4. How is rain formed?

Answer: Water evaporates from the Earth's surface and rises into the atmosphere. It cools and condenses into tiny droplets. These droplets join to form larger drops. When the drops become heavy, they fall as rain.

Q5. What is the water cycle?

Answer: The water cycle is the continuous movement of water between the Earth's surface and the atmosphere through evaporation, condensation, and precipitation.

Q6. What are the main steps of the water cycle?

Answer: The main steps are:

1. Evaporation: Liquid water changes into water vapour.
2. Condensation: Water vapour cools and forms tiny water droplets.
3. Precipitation: Water falls to the Earth as rain, hail, or snow.
4. Collection: Water collects in rivers, lakes, oceans, and the ground.

Q7. How does the water cycle help living organisms?

Answer: The water cycle returns water to the Earth's surface and helps maintain the supply of water needed by plants, animals, and humans.

Q8. How does water return to the Earth from clouds?

Answer: Water returns to the Earth as precipitation in the form of rain, hail, or snow.

12. Water Conservation

Q1. What is water conservation?

Answer: Water conservation means using water carefully and preventing its wastage.

Q2. Why is water conservation important?

Answer: Water conservation is important because only a small amount of Earth's water is easily available freshwater. We must save it for present and future generations.

Q3. How can we conserve water in our daily lives?

Answer:

- Turn off taps when not in use.
- Repair leaking taps and pipes.
- Use only the water we need.
- Reuse water when it is safe to do so.
- Collect rainwater where possible.
- Avoid polluting rivers, lakes, and ponds.

Q4. How does water pollution affect the availability of water?

Answer: Pollution makes water unsafe for drinking and other uses. It reduces the amount of usable water available to us.

Q5. Why should waste from homes, industries, and farms not be thrown into water bodies?

Answer: Such waste can pollute water and make it unfit for consumption. It can also harm aquatic life.

Q6. How do trees help in water conservation?

Answer: Trees help rainwater enter the soil and reduce surface runoff. They also release water vapour through transpiration and are an important part of the water cycle.

13. Additional Important Questions and Answers

These questions cover the key concepts of the chapter and are useful for revision and examinations.

Q1. What is the difference between evaporation and condensation?

Answer: Evaporation changes a liquid into vapour, while condensation changes vapour into a liquid.

Q2. What is the difference between melting and freezing?

Answer: Melting changes a solid into a liquid by gaining heat. Freezing changes a liquid into a solid by losing heat.

Q3. Why does water evaporate faster in summer than in winter?

Answer: The temperature is usually higher in summer, which increases the rate of evaporation.

Q4. Why does a wet floor dry faster under a fan?

Answer: Moving air carries water vapour away from the floor and increases the rate of evaporation.

Q5. Why does a puddle of water disappear after a sunny day?

Answer: The Sun heats the water, causing it to evaporate into water vapour.

Q6. Why does a cold glass become wet on the outside?

Answer: Water vapour in the air condenses on the cold outer surface of the glass.

Q7. Can water vapour be seen directly?

Answer: Water vapour is an invisible gas. The white cloud seen above boiling water consists mainly of tiny liquid water droplets formed when vapour cools.

Q8. Why is the water cycle called a continuous process?

Answer: Water continuously evaporates, condenses, and returns to the Earth through precipitation.

Q9. What happens to water molecules when water is heated?

Answer: The molecules gain energy and move faster. With sufficient heating, liquid water changes into water vapour.

Q10. What happens to water molecules when water is cooled?

Answer: The molecules lose energy and move more slowly. Depending on the temperature, water vapour may condense into liquid water, or liquid water may freeze into ice.

Q11. What is the difference between gas and vapour according to the chapter?

Answer: A gas is a substance that exists in the gaseous state under normal conditions. A vapour is the gaseous form of a substance that is normally solid or liquid under those conditions.

Q12. What is the role of the Sun in the water cycle?

Answer: The Sun provides heat energy that causes water to evaporate from rivers, lakes, seas, and other water bodies.

14. Multiple-Choice Questions (MCQs)

Practice Quiz

0/12 answered

1. Which percentage of the Earth's surface is covered by water?

A. 21%
B. 51%
C. 71%
D. 91%

2. In which state does water have a fixed shape and fixed volume?

A. Solid
B. Liquid

C. Gas
D. Vapour only

3. The change of ice into water is called:

A. Freezing
B. Melting
C. Condensation
D. Evaporation

4. The change of water into ice is called:

A. Boiling
B. Melting
C. Freezing
D. Condensation

5. Which process changes a liquid into vapour?

A. Condensation
B. Vaporisation
C. Freezing
D. Melting

6. Evaporation takes place:

A. Only at 100°C
B. Only in winter
C. At all temperatures
D. Only in sunlight

7. Which factor decreases the rate of evaporation when it increases?

A. Temperature
B. Wind speed
C. Surface area
D. Humidity

8. Why does water in a clay pot become cool?

A. The pot produces ice
B. Water evaporates through its pores
C. The pot absorbs sunlight
D. The water boils

9. The formation of water droplets from water vapour is called:

A. Melting
B. Freezing
C. Condensation
D. Boiling

10. Water falling from clouds as rain, hail, or snow is called:

A. Evaporation
B. Transpiration
C. Precipitation
D. Collection

11. Water loss from plant leaves as water vapour is called:

A. Condensation
B. Transpiration
C. Freezing
D. Boiling

12. Which source of energy drives evaporation in the water cycle?

A. Moon
B. Sun
C. Stars
D. Soil
Check Answers

15. Fill in the Blanks

Fill in the blanks and then check your answers.

1. About ___% of the Earth's surface is covered by water.

2. The solid state of water is called ___.

3. The change of a solid into a liquid is called ___.

4. The change of a liquid into a solid is called ___.

5. The change of a gas into a liquid is called ___.

6. Water loss from plant leaves as vapour is called ___.

7. The amount of water vapour in the air is called ___.

8. Higher temperature generally ___ the rate of evaporation.

9. Higher humidity generally ___ the rate of evaporation.

10. Rain, hail, and snow falling from clouds are forms of ___.

Show Answers

16. True or False

1. Water exists in three states. — True
2. Ice has no fixed shape. — False
3. Evaporation occurs only at the boiling point. — False
4. Wind can increase the rate of evaporation. — True
5. High humidity increases the rate of evaporation. — False
6. Condensation changes vapour into liquid. — True
7. The Sun helps drive the water cycle. — True
8. Water in a clay pot may become cool due to evaporation. — True
9. Water pollution makes water safer for drinking. — False
10. Water conservation is important for future generations. — True

17. Quick Revision Notes

- Water: An essential natural resource for all living organisms.
- Three states: Solid (ice), liquid (water), and gas (water vapour).
- Melting: Solid → Liquid.
- Freezing: Liquid → Solid.
- Vaporisation: Liquid → Gas or vapour.
- Evaporation: Change of liquid into vapour at the surface; it can occur at all temperatures.
- Boiling: Rapid change of liquid into vapour at its boiling point.
- Condensation: Gas or vapour → Liquid.
- Transpiration: Loss of water vapour from plants.
- Rate of evaporation: Increases with greater surface area, higher temperature, and faster wind; decreases with higher humidity.
- Cooling effect: Evaporation takes heat from the surroundings.
- Water cycle: Evaporation → Condensation → Precipitation → Collection.
- Water conservation: Save water and prevent pollution.

Note: The uploaded PDF contains pages 104–111 of the chapter. It does not include the chapter's final exercise pages, so the answers above cover all the questions visible in the uploaded pages, along with additional topic-wise practice questions.

08-a-journey-through-the-state-of-water.pdf